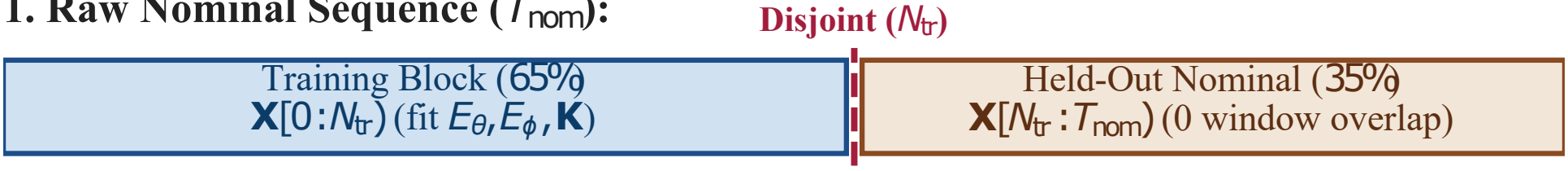
## 1. Raw Nominal Sequence ($T_{nom}$):




## 2. Held-Out Causal Score Indexing ($W=C+S=320$, $\tau=t+S-1$):




At gap $S=64$: suspect targets $\mathbf{x}_{tgt}$ are 100% disjoint ($W=320$ check: $\Delta\text{FPR} \le 0.002$).

## 3. Online Window Causality ($\vec{\tau}$): Emission Lag vs. Detection Delay:

